

Calendar:

Local system:

- Security policies
- Access control (what is the user allowed to do on the system)
- DB Security
- Software Security (correctness of software programs)
- Operating System security
- Trusted computing

Interact with other systems (information flow) $\Leftrightarrow$

Authentication over the internet

Internet security protocols

Wireless sys. sec.

Distributed sys. Sec.

Cloud sec.

Copyright sec.

Privacy prot.

Info. hiding

Attacks and defenses (bad actors):

- malware
- Ransomware
- ddos
- (defenses):
- Intrusion detection
- Intrusion Protection
- Physical tamper resistance
- emission security

Management Issues

- risk assessment
- Security controls (2-person approval)
- Physical security (locked doors)
- Security awareness

Alice	(Eavesdropper (Eve)) (passive, active)	Bob
	(transmitted over channel)	

$m = (111011)$	(111011)
$E_k(m) = c$	$D(c) = m$

$K=K'$: symmetric cipher

Confidentiality from the eavesdropper